FE BAR SUBMISSION · BUSINESS VALUE BRIEF

# Hire Right

## AI-powered predictive hiring for Jackson & Jackson HR Digital

From raw applicant data to a governed, conversational hiring decision — one connected journey on Databricks.

THE PROBLEM

# Hiring senior HR leaders is slow, subjective, and expensive to get wrong.

- **45+ days** to fill a director-level role — top candidates are gone before the shortlist is set.
- Decisions rest on **gut feel and inconsistent scorecards**, creating bias and compliance exposure in a regulated industry.
- A single **mis-hire at the leadership level costs 1.5–3× salary** in severance, lost productivity, and re-recruiting.
- Applicant data, resumes, and analytics live in **disconnected tools** — no single, governed source of truth.

THE OUTCOME — LEAD WITH IMPACT

# Faster, fairer, defensible hiring decisions.

**60%**
**Faster time-to-fill**
Ranked shortlists on day one, not week six

**~30%**
**Fewer mis-hires**
8-factor ML scoring lifts quality of hire

**15 hrs**
**Saved per requisition**
Automated screening & resume Q&A

**100%**
**Auditable & consistent**
Governed scoring, masked PII, tracked notes

Projected impact for the J&J HR Digital prototype, expressed in the KPIs the CHRO and TA leadership already report on.

WHAT THE HR TEAM GETS

# A decision cockpit for every requisition.

### Ranked candidates
Every applicant scored across 8 weighted dimensions with a clear total & stage.

### Data-science recommendation
One click runs the ML prediction and returns a hire recommendation with rationale.

### Resume Q&A (RAG)
Ask about any candidate's experience — grounded in their actual resume.

### Genie analytics
"Who are the top 3 for the CPO role?" answered instantly over governed data.

### HR notes
Add durable, audit-trailed annotations to a candidate record — saved to Lakebase.

### AI offer letter
One click drafts a salaried offer letter and sends it via the agent's mailer tool.

HOW IT WORKS

# One connected journey — data to decision on Databricks.

01 · FOUNDATION

### Lakeflow + Unity Catalog

Synthetic applicant data ingested with Lakeflow, refined Bronze→Silver→Gold, governed with lineage, PII masking & tags.

02 · SERVE

### Lakebase

Governed tables synced to a low-latency Postgres layer that powers the app — and stores HR annotations.

03 · PREDICT

### Predictive ML

An MLflow hire model on Model Serving returns a hire / no-hire decision from the scoring features.

04 · AGENT

### AI Agent

A ResponsesAgent orchestrates tools — Genie, resume RAG, ML scoring, and email — grounded in your data.

05 · EXPLORE

### Genie Room

Recruiters ask questions in natural language over governed data — no SQL required.

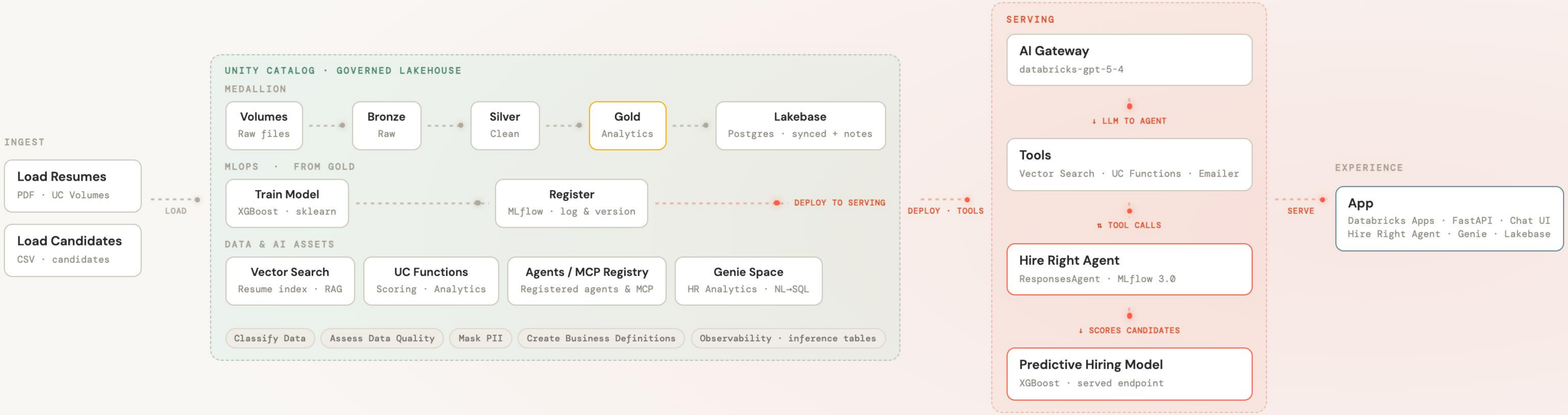
06 · DELIVER

### Databricks App

A single HR workspace: scoring, recommendations, resume viewer, Genie Q&A, and candidate notes.

ARCHITECTURE · KEY TECHNICAL COMPONENTS

# End-to-end on Databricks — data to decision.




→ data / build flow ⇄ / ⇅ request & response coral = serving runtime · grey = pipeline

TECHNICAL · DATA FOUNDATION

# A governed medallion, ingested with Lakeflow.

INGEST

### Lakeflow

Synthetic resumes + candidate/job CSVs land in UC Volumes and are read incrementally with Lakeflow (`read_files` / Auto Loader).

BRONZE

### Raw

As-ingested, full-fidelity and replayable — the immutable source of truth.

SILVER

### Clean & governed

Typed candidates & job requirements; PII masked; data-quality checks applied.

GOLD

### Analytics-ready

`candidate_scoring_summary` — joined, scored features powering ML, Genie & the app.

TRUST & GOVERNANCE

# Built for a regulated industry from day one.

### PII protected

Column masking & sensitivity tags on candidate data enforced in Unity Catalog.

### Full lineage

Every field traceable Bronze→Gold→Lakebase for audit and explainability.

### Evaluated AI

The agent is scored with MLflow evaluation and an LLM-judge before it ships.

### Accountable notes

Every HR annotation is attributed and time-stamped in a transactional store.

The same governance that protects candidates also gives the CHRO a defensible, explainable hiring process.

TECHNICAL · OPERATIONAL SERVING

# Lakebase turns the lakehouse into an app backend.

### Synced tables

Gold candidate & job tables synced Delta→Postgres for millisecond, app-facing reads.

### Transactional notes

A native `candidate_annotations` table captures HR notes, joined to candidate data.

### Powers the app

Candidate cards, job views and notes all served from Lakebase — a snappy UX.

### Secure by default

The app connects with short-lived OAuth credentials, governed in Unity Catalog.

TECHNICAL · PREDICTIVE ML

# From labeled history to a served hire model.

**01 · TRAIN**

### Three models

LogReg, Random Forest & Gradient Boosting on the 8 scoring features (Silver).

**02 · TRACK**

### MLflow

Every run logged — accuracy, AUC, F1 and feature importances, fully comparable.

**03 · REGISTER**

### @champion

The best model is promoted to `@champion` in Unity Catalog.

**04 · SERVE**

### Model Serving

A scale-to-zero real-time endpoint returns the hire / no-hire decision.

TECHNICAL · AGENTIC AI

# A ResponsesAgent that reasons with your data.

**query_genie**
Natural-language analytics over the governed Gold layer.

**search_resumes**
Vector Search RAG grounded in candidate resumes.

**predict_hiring_score**
Calls the served ML model for a hire recommendation.

**send_email**
Mailgun-backed mailer — powers the AI offer letter.

MLflow 3.0 ResponsesAgent with a Python tool-calling loop — and tool calls **stream live into the app** as they run, so HR sees the agent's work in real time.

TECHNICAL · SELF-SERVE ANALYTICS

# Genie — ask the data in plain English.

- **NL→SQL** over the governed Gold layer, guided by business semantics & metric definitions.
- Powers both the app's **Genie panel** and the agent's **query_genie** tool.
- No SQL required — recruiters get instant answers that stay **governed and auditable**.

TECHNICAL · TRUSTWORTHY AI

# Quality gates before the agent ships.

### MLflow evaluation
Built-in and custom scorers run against a curated eval set.

### LLM-as-judge
An LLM judge scores tone & consistency on every response.

### Tracing
Each tool call is traced for debugging and production observability.

### Gated deploy
Only agents that clear the quality bar are promoted to serving.

WHY DATABRICKS

# Data, AI, and the app — on one governed platform.

- **No integration tax:** ingestion, governance, ML, Gen AI, analytics, and the app all live together.
- **Lakebase** turns governed lakehouse tables into a low-latency application backend — no separate database to sync and secure.
- **Unity Catalog** governs data *and* AI, so the whole solution is auditable end-to-end.
- **Faster to value:** this entire prototype was built as one connected journey, not stitched from point tools.

THE ASK

# Let's hire right, together.

A working, governed, AI-powered hiring solution — ready to pilot on a live requisition and prove the numbers on your data.

- Pilot on one open role and baseline time-to-fill & quality of hire
- Extend the model and Genie room to your historical hiring data
- Roll the app out to the TA team behind Unity Catalog governance